

specific tasks or motions while a robot takes input from its surrounding, processes the data, and then interacts with the environment. A robot can be programmed to perform a variety of tasks and is capable of making decisions about its actions based on a number of factors. It might seem surprising to some, but there is still debate about the exact definition of what a robot is and there's a lot of ambiguity surrounding the word. A toaster is a simple machine because it is capable of performing only a certain task and does not take decisions of its own, while a 3D printer can be called a robot, as it takes input about the design and then processes it to determine how exactly to print the model. Not every machine can be called a robot while every robot will have some or the other element of these simple machines.

## Characteristics of a robot

A machine needs to have some defining characteristics before being counted as a robot. Following are the basic defining characteristics of a robot that a machine must have to be called a robot:

- **Sense** – The machine should be capable of interacting with its environment and sensing its surrounding. This can be done using any number of sensors such as light sensors, chemical sensors, pressure and touch sensors etc. The machine needs to know about its surrounding in order to perform an action.
- **Movement** – Either the machine should be capable of movement in its surrounding environment or it should have moving parts that somehow interact with the environment. This part of the robot is called 'actuators'.
- **Energy** – The machine should have a power source that provides it with energy. It can be anything from solar power to rechargeable batteries.
- **Intelligence** – The most defining characteristic of a robot, intelligence is what primarily sets apart a robot from a machine. It should possess some sort of intelligence such that it processes and takes decisions about its action based on the data it collects from its sensors. Unlike popular misconception, a robot doesn't have to necessarily look like us humans. In fact, most of today's robots don't. An

